

Shujia Zhang

jean.zhang@yale.edu | jeanzsj.github.io | new haven & shanghai

Education

Yale University

Expected Grad: May 2029

B.S. in Physics.

Relavant Coursework: Classical Mechanics, Relativity, Quantum Information Processing, Linear Algebra, Introduction to Computer Science, Introduction to Computer Engineering, Modern Physical Measurement.

Skills

Programming: Python

Quantum Computing: basic quantum information & quantum algorithms (e.g., Grover's, Shor's); familiarity with Qiskit (IBM), Guppy & Selene (Quantinuum's language and emulator).

Hardware & Systems: fundamentals in computer architecture; implementing digital logic in Verilog with Vivado; simulated and tested on FPGA platforms; built and prototyped embedded systems using Arduino.

Instrumentation: familiar with experimental design, data analysis and error analysis; familiar with circuitry and measurement devices such as oscilloscopes.

Astronomy & Data Analysis: familiar with telescope operation and data processing with PixInsight.

Web Development: HTML, CSS

Projects

Benchmarking QAOA Against Classical Optimization Method

Apr 2026

Implemented QAOA (Quantum Approximate Optimization Algorithm) in Python and Guppy, ran on Quantinuum's emulator Selene. Benchmarked its speed and accuracy against the classical method (PuLP CBC). Won 3rd place in YQuantum Hackathon - Travelers-LTImindTree-Quantinuum-Challenge.

Implementing Shor's Algorithm With Qiskit

Apr 2026

Implemented Shor's Algorithm for factoring 15 with Qiskit. Discovered and reported bug in Qiskit's Quantum Shannon Decomposition (certain permutation matrices hit numerically unstable cases in the Schur decomposition method). Resolved with using Isometry to force an alternate decomposition method.

Blog

Jan 2026-Present

Posts course notes, relevant toolkits, and blogs regularly at jeanzsj.github.io

Research

Summer Science Program

GCSU, GA · Jun 2024 – Jul 2024

Researched asteroid orbits & Saturn's influence on orbital periodicity; published on Minor Planet Circulars.

Pioneer Research Program – Relativity & Cosmology

Online · Feb 2024 – Jul 2024

Researched the feasibility of black hole energy extraction through the Penrose Process to address the energy crisis.

Astronomy Research @ SJTU

Shanghai, China · Oct 2023 – Present

Simulated gravitational lensing and calculated cosmic shears.

Stanford Summer Humanities Institute

Palo Alto, CA · Jul 2023 – Jul 2023

Researched how Marquez grants Macondo chance to combat solitude through creating a common identity.